

Homework 1 - Set Notation, Counting, and Probability Axioms

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. (8 points) A box contains 3 orange, 4 purple, 5 silver, and 6 black balls. A sample of 5 balls is chosen without replacement. How many **possible samples** of size 5 include all four colors?

Answer: To include all four colors in a sample of size 5, one color must appear twice and the other three colors must appear once.

$$\begin{aligned} & \binom{3}{2} \binom{4}{1} \binom{5}{1} \binom{6}{1} + \binom{3}{1} \binom{4}{2} \binom{5}{1} \binom{6}{1} + \binom{3}{1} \binom{4}{1} \binom{5}{2} \binom{6}{1} + \binom{3}{1} \binom{4}{1} \binom{5}{1} \binom{6}{2} \\ &= 360 + 540 + 720 + 900 \\ &= 2520 \end{aligned}$$

2. (5 points) A security badge number consists of two uppercase letters followed by three digits. The two letters may not be repeated, the first digit cannot be 0, and the remaining digits may be repeated. How many **different** badge numbers are possible?

Answer: There are 26 choices for the first letter and 25 choices for the second letter.

There are 9 choices for the first digit, then 10 choices for each of the remaining two digits.

$$\text{Total} = 26 \times 25 \times 9 \times 10 \times 10 = 585000.$$

3. (5 points) A committee of 4 people is selected from 6 faculty members and 5 graduate students. What is the **probability** that the committee contains exactly 2 faculty members and 2 graduate students?

Answer: Total committees: $\binom{11}{4}$

Favorable committees: $\binom{6}{2} \binom{5}{2}$.

$$P = \binom{6}{2} \binom{5}{2} / \binom{11}{4} = (15)(10)/330 = 5/11.$$

4. (7 points) What is the **probability** that a hand of 5 cards chosen randomly and without replacement from a standard deck of 52 cards contains the king of spades, exactly 1 other king and exactly 2 queens?

Answer: The hand must include the king of spades, exactly 1 of the remaining 3 kings, exactly 2 of the 4 queens, and 1 other card that is neither a king nor a queen.

Favorable hands: $\binom{3}{1} \binom{4}{2} \binom{44}{1}$

Total hands: $\binom{52}{5}$.

$$P = \binom{3}{1} \binom{4}{2} \binom{44}{1} / \binom{52}{5} = 792/2598960 \approx 0.000305.$$

5. (6 points) An urn contains 4 red balls, 3 blue balls, and 3 green balls. Four balls are selected at random **without replacement**. What is the probability that at least one color is not selected?

Answer: Use the complement: $P(\text{at least one color not selected}) = 1 - P(\text{all three colors selected})$.

Total samples: $\binom{10}{4} = 210$.

To select all three colors in 4 draws, one color appears twice and the other two colors appear once:

$$\binom{4}{2}\binom{3}{1}\binom{3}{1} + \binom{4}{1}\binom{3}{2}\binom{3}{1} + \binom{4}{1}\binom{3}{1}\binom{3}{2} = 54 + 36 + 36 = 126.$$

$$P = 1 - 126/210 = 84/210 = 2/5.$$

6. (6 points) An urn contains 10 balls: 5 are white, 3 are red and 2 are black. Three balls are drawn at random, **with replacement**, from the urn. What is the probability that all 3 balls are different colors?

Answer: All three draws must be one white, one red, and one black in any order.

$$P = (5 \times 3 \times 2 \times 3!) / 10^3 = 180/1000 = 0.18.$$

7. (9 points) A shelf contains 5 **different** statistics books, 4 **different** algebra books, and 3 **different** calculus books. How many ways can the 12 books be arranged if all statistics books must be together and all algebra books must be together? Assume all books are distinct.

Answer: Treat all statistics books as one block and all algebra books as one block.

Then arrange 5 objects: the statistics block, the algebra block, and the 3 distinct calculus books. This gives $5!$ arrangements.

Within the statistics block, the books can be arranged in $5!$ ways. Within the algebra block, the books can be arranged in $4!$ ways.

$$\text{Total} = 5! \times 5! \times 4! = 120 \times 120 \times 24 = 345600.$$

8. (10 points) A fair six-sided die is rolled **five** times. What is the probability of obtaining exactly two 6s and exactly one odd number among the remaining three rolls?

Answer: Choose the positions of the two 6s: $\binom{5}{2}$.

Among the remaining 3 positions, choose one position for an odd non-6 result: $\binom{3}{1}$. The possible odd values are 1, 3, and 5, so there are 3 choices.

The remaining two positions must be even non-6 values, namely 2 or 4, so there are 2×2 choices.

$$\text{Favorable outcomes} = \binom{5}{2}\binom{3}{1} \times 3 \times 2 \times 2 = 360.$$

$$\text{Total outcomes} = 6^5 = 7776.$$

$$P = 360/7776 = 5/108.$$

9. (6 points) A class has 18 first-year students and 12 second-year students. Five students are selected at random without replacement. What is the probability that at least 4 of the selected students are second-year students?

Answer: At least 4 second-year students means either 4 or 5 second-year students.

$$P = \left[\binom{12}{4} \binom{18}{1} + \binom{12}{5} \right] / \binom{30}{5} \\ = [495(18) + 792] / 142506 = 9702/142506 = 77/1131.$$

10. (8 points) Prove or disprove the following set identity: $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$.

Answer: Left side: $A \cap (B \setminus C) = A \cap (B \cap C^c) = A \cap B \cap C^c$.

Right side: $(A \cap B) \setminus (A \cap C) = (A \cap B) \cap (A \cap C)^c$.

Using De Morgan's Law, $(A \cap C)^c = A^c \cup C^c$.

Thus, $(A \cap B) \cap (A^c \cup C^c) = (A \cap B \cap A^c) \cup (A \cap B \cap C^c)$.

Since $A \cap A^c = \text{empty set}$, the first part is empty, leaving $(A \cap B \cap C^c)$.

Both sides are equal, so the identity is true.

11. (9 points) Suppose $P(A) = 0.55$, $P(B) = 0.40$, and $P(A \cup B) = 0.75$. Find $P(A \cap B)$, $P(A \cap B^c)$, and $P(A^c \cap B^c)$.

Answer: Use $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

$$0.75 = 0.55 + 0.40 - P(A \cap B), \text{ so } P(A \cap B) = 0.20.$$

$$P(A \cap B^c) = P(A) - P(A \cap B) = 0.55 - 0.20 = 0.35.$$

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.75 = 0.25.$$

12. (5 points) Suppose $P(A) = 0.60$, $P(B) = 0.50$, $P(C) = 0.40$, $P(A \cap B) = 0.25$, $P(A \cap C) = 0.20$, $P(B \cap C) = 0.15$, and $P(A \cap B \cap C) = 0.10$. Find the probability that at least one event occurs.

Answer: Use inclusion-exclusion:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C) \\ = 0.60 + 0.50 + 0.40 - 0.25 - 0.20 - 0.15 + 0.10 = 1.00.$$

13. (7 points) A box contains 8 good components and 3 defective components. Four components are selected at random without replacement. What is the probability that **at most** one selected component is defective?

Answer: At most one defective means either 0 defective or 1 defective.

Total samples: $\binom{11}{4}$

$$\text{Favorable samples: } \binom{8}{4} + \binom{8}{3} \binom{3}{1} = 70 + (56)*3 = 238.$$

$$P = 238/330 = 119/165.$$

- 14. (9 points)** A fair six-sided die is rolled **twice**. What is the probability that
- a) a six turns up exactly once?
 - b) both numbers are odd?
 - c) the sum of the scores is 4?

Answer: There are $6^2 = 36$ total outcomes.

(a) Exactly one six: (6, not 6) or (not 6, 6). There are $5 + 5 = 10$ favorable outcomes. $P = 10/36 = 5/18$.

(b) Both numbers are odd: the odd numbers are 1, 3, and 5. There are $3 \times 3 = 9$ favorable outcomes. $P = 9/36 = 1/4$.

(c) Sum is 4: favorable outcomes are (1,3), (2,2), and (3,1). $P = 3/36 = 1/12$.

Final answer: (a) $5/18$, (b) $1/4$, (c) $1/12$